

Implementing Verdier's robust extrapolation in the CKT model

Relaxing the i.i.d. migration-cost assumption

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Our current estimator is Verdier's “simple” extrapolation

What we currently do. Following Suri (2011) and Tjernström (2023):

- impose the LCA restriction $\Delta_i = \beta + \phi \theta_i$;
- identify ϕ from switcher trajectories;
- extrapolate to non-migrants (d_N, d_T) via the LCA line.

What Verdier (2020) shows. This procedure is one specific instance of a broader family. He calls our version the **simple extrapolation** and proposes a **robust** alternative that weakens one specific assumption.

What we propose. Adopt Verdier's robust version.

The simple extrapolation assumes i.i.d. migration-cost shocks

Our current Assumption A2 states that the non-pecuniary utility shock

ν_{it}^l is i.i.d. across individuals i , time t , and locations l .

What this implies. The residual migration-cost shifter is independent of baseline skill and comparative advantage (τ_i, θ_i) **everywhere in the sample**.

What this rules out. Any systematic difference in the distribution of migration-cost shocks across regions, institutions, or subpopulations.

Practical reading

Under A2, two workers in completely different parts of the country face the same distribution of migration cost, conditional only on their own comparative advantage.

The migration literature documents exactly what A2 rules out

Regional variation in migration cost is the central fact of this literature:

- Bryan and Morten (2019): internal migration costs in Indonesia large enough to move Indonesia toward the world aggregate misallocation frontier.
- Lagakos et al. (2020): migration costs vary substantially within and across developing countries.
- Pulido and Swiecki (2021): sector-side labor-market frictions correlate with regional productivity.
- China specifically: hukou institutions generate large, regionally-varying barriers to rural-to-urban migration.

Our current model rules this out by assumption.

Verdier's robust extrapolation weakens A2 to a conditional version

Proposal. Decompose the migration-cost shock into a region-specific component and a residual:

$$\nu_{it}^l = m_l(v_i) + \tilde{\nu}_{it}^l,$$

where v_i is an observed, time-invariant individual-level index.

New assumption (A2'). Only the residual $\tilde{\nu}_{it}^l$ is i.i.d. conditional on v_i . The region-level component $m_l(v_i)$ can be arbitrarily correlated with baseline skill and comparative advantage.

What we pick for v_i . First-wave province (same across all three countries, to start).

Why this is weaker

A2 required independence of ν_{it}^l from (τ_i, θ_i) unconditionally. A2' requires only that the within-province variation be independent.

The LCA gets a province-specific intercept; slope stays common

Old LCA (simple extrapolation):

$$\Delta_i = \beta + \phi \theta_i.$$

New LCA (robust extrapolation):

$$\Delta_i = \beta(v_i) + \phi \theta_i.$$

- Slope ϕ is **common across provinces** — this remains the LCA restriction.
- Intercept $\beta(v_i)$ absorbs province-level migration-cost heterogeneity.
- Identification of ϕ comes from switcher variation within a province.
- Extrapolation to non-migrants happens province-by-province, then averaged.

Pooling provinces mixes differentially-selected switcher samples

Consider two provinces:

- Low-cost province (coastal, good infrastructure). Migration cost low. Many types migrate. Switcher sample covers a wide range of θ_i .
- High-cost province (remote, poor infrastructure). Migration cost high. Only unusually high-return types migrate. Switcher sample is skewed.

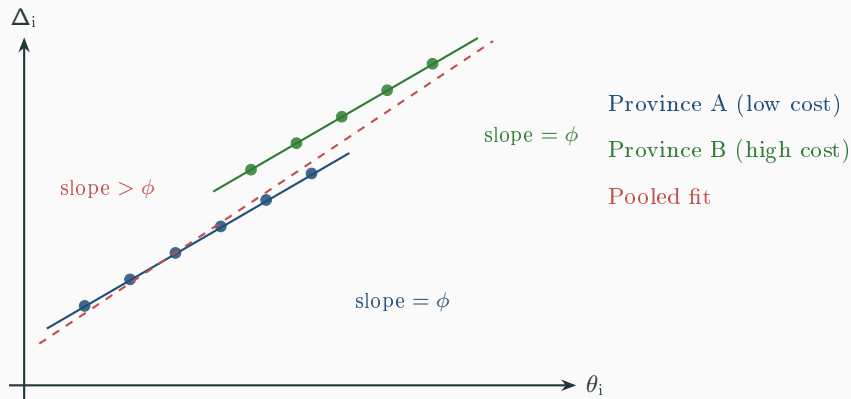
Pooled regression of Δ_i on θ_i : biased by the across-province difference in selection.

Within-province regression: restricts identification to comparisons where migration cost is held constant. Clean slope.

This is what partialling out v_i achieves

It purges the selection bias induced by region-level migration-cost heterogeneity.

Within-province slopes agree; the pooled slope is pulled off



Each province's switcher cloud has the true LCA slope ϕ . The pooled line is tilted by the province-level intercept difference, not by a real change in ϕ .

Implementation adds v_i -interacted means to the GRC regression

The restricted GRC, currently:

$$y_{it} = \sum_{\underline{d}} \mu_{\underline{d}} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \Delta_{\underline{d}_0} D_{it} + \sum_{\underline{d} \in \mathcal{D}_S \setminus \underline{d}_0} \phi(\mu_{\underline{d}} - \mu_{\underline{d}_0}) D_{it} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \cdots + \varepsilon_{it}.$$

Under the robust version: trajectory means become trajectory $\times$ province means.

$$y_{it} = \sum_{\underline{d}, v} \mu_{\underline{d}, v} \mathbb{1}\{\underline{d}_i = \underline{d}, v_i = v\} + \Delta_{\underline{d}_0} D_{it} + \phi \sum_{\underline{d} \neq \underline{d}_0} (\mu_{\underline{d}, v_i} - \mu_{\underline{d}_0, v_i}) D_{it} \mathbb{1}\{\cdot\} + \cdots$$

- Slope ϕ unchanged — still a single scalar.
- Standard errors cluster at v_i .
- Stayer return $\Delta_{\underline{d}_N}$ computed within each v_i -cluster, then averaged over clusters with switcher support.

The cost-shifter diagnostic is a falsification test, not a substitute

Verdier also proposes a **separate** diagnostic, using an observed continuous cost shifter w_i :

$$\theta_i^* = \alpha_0 + \alpha_1 \Delta_i + \alpha_2 w_i + e_i, \quad H_0 : \alpha_2 = 0.$$

In Kenya he uses w_i = average distance to nearest seed seller.

- Requires an observed measured exogenous cost shifter.
- Purpose: falsify the simple extrapolation. If $\alpha_2 \neq 0$, the simple assumption is wrong.
- Does not deliver an alternative estimate on its own.
- Not required to implement the robust extrapolation; the robust version uses v_i instead.

For now

We focus on the robust extrapolation via v_i = province. The cost-shifter test is a natural robustness addition if we can identify a defensible w_i per country.

The move buys a weaker assumption; the costs are tractable

What we gain.

- A2' is substantially weaker than A2 — it matches what the migration literature documents.
- The CHN hukou split is now a principled special case (with $v_i = \text{hukou}$) rather than an ad hoc fix.
- Stayer ATE estimates are defensible against the concern Verdier raised in Suri's own data.

What we give up.

- Δ_{d_N} is now an average over provinces with switcher support, not the full population average. (Feasibility survey will report support fractions.)
- Thinner identification of ϕ : only within-province switcher variation contributes.
- Cluster-robust standard errors at the v_i level.

Feasibility survey first, then implementation

In progress. Read-only survey of **data/processed/** for each country:

- Which geographic variables are available at baseline.
- Number of unique values at each granularity (province, district, community).
- Switchers per cluster; always-rural support fraction per cluster.

Next. Implement v_i = first-wave province across all three countries in **run_grc**.

Then. Report simple-extrapolation and robust-extrapolation estimates side by side; run Verdier's overidentification test of the LCA line to diagnose any remaining misfit.

Punt for later. Check whether the raw CFPS/IFLS/TZNPS files record province of origin or province of birth; if so, substitute for first-wave province in a robustness specification.